

Abdul Ali

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Professional Summary

Ph.D. candidate in Energy Science and Engineering with expertise in electrochemical energy storage systems, specifically vanadium redox flow batteries. Over 6 years of research experience in physics-based battery modelling, simulation, and optimization during my PhD. Developed advanced differential-algebraic equation (DAE) solver techniques and performed extensive COMSOL Multiphysics simulations to enhance battery performance. Passionate about sustainable energy R&D, with a strong blend of technical skills and leadership experience in academic and entrepreneurial initiatives.

Education

- **Ph.D. in Energy Science and Engineering**, Indian Institute of Technology Bombay (2018 – Present, Expected 2025). *Thesis*: “Physics-Based Modelling and Analysis of Electrode Parameters for Enhanced Vanadium Redox Flow Battery Performance.” Focus on redox flow battery modelling & simulation and high-index DAE solver development.
- **M.Sc. in Energy Science and Engineering**, IIT Bombay (2015 – 2017) I specialize in initialization and numerical solutions for complex DAE systems. I completed a research project on one-step solution methods for DAE-based battery models.
- **B.Sc. in Physics, Chemistry & Mathematics**, CSJM University, Kanpur (2012 – 2015)
- **Higher Secondary Education (Standard XII)**, SSB Inter College, Deoria, Uttar Pradesh, India (July 2010 – June 2012)
 - Subjects: Physics, Chemistry, Mathematics, Hindi, English
- **Secondary Education (Standard X)**, SLJ Inter College, Deoria, Uttar Pradesh, India (July 2008 – June 2010)
 - Subjects: Science, Mathematics, Social Science, English, Hindi, Sanskrit

Research Experience

- **Ph.D. Research Scholar, IIT Bombay (2018 – Present):** Developed a 2D physics-based model of a vanadium redox flow battery (VRFB) to analyze the impact of electrode microstructure (fibre diameter and porosity) on battery performance. Improved DAE solver methodologies (e.g., singular perturbation-based initialization and one-step solution approach) to handle the stiff coupled electrochemical equations in VRFB simulations. Conducted extensive modelling in MATLAB/Simulink and COMSOL, providing insights that ~90% electrode porosity and optimized fibre size 40-70 μ m range significantly improve system efficiency while minimizing pumping losses. This work led to multiple conference presentations and journal manuscripts (under review) on VRFB optimization.
- **Development and Delivery of Iron Redox Flow Battery (2022):** Designed, fabricated, and delivered a 5-watt all-iron redox flow battery prototype to **Savitribai Phule Pune University (SPPU), School of Energy Studies**. The system comprised two 5-cell stacks—one fully assembled and one in kit form. Oversaw material selection,

integration of flow field components (carbon cloth, membranes, graphite and brass plates), and assembly protocols. Provided comprehensive documentation and supporting hardware including tanks, connectors, and tubing. This work bridged modeling insights with hands-on fabrication, demonstrating successful academic-to-practical technology transfer in redox flow battery prototyping.

Technical Skills

- **Programming & Data Analysis:** MATLAB, Python, Maple, Mathematica (experienced in numerical modeling, data analysis, and simulation scripting)
- **Simulation & Modeling Tools:** COMSOL Multiphysics (electrochemical and multiphysics simulations), MATLAB/Simulink, Mathematica (analytical computations)
- **Web Development:** HTML, CSS, WordPress, Drupal (website design, content management, and deployment)
- **Other Software:** Adobe Photoshop (graphic design), Microsoft Office Suite

Projects

- **Indo-UK Energy (IUCERCE) Project, IIT Bombay (2019 – 2022):** Contributing to the design of an independent solar-plus-battery powered microgrid. Involves system modelling of solar PV and vanadium flow batteries, optimizing integration for reliable off-grid power in rural communities.
- **DAE Solver Development (M.Sc. Project), IIT Bombay (2016 – 2017):** Developed a singular perturbation-based initialization and one-step solution method for high-index DAE systems, improving convergence and stability for complex energy system simulations. Applied the solver to redox flow battery models as a proof of concept.
- **Unitized Regenerative Fuel Cell Study, IIT Bombay (Jan – Jun 2016):** Analyzed the operation and efficiency of a unitized regenerative fuel cell (URFC) system capable of energy storage and generation. Evaluated performance metrics during charging/discharging modes as part of a sustainable energy project.

Professional Experience

- **Executive Councilor, Research Scholars Forum (RSF), IIT Bombay (Sept 2019 – Aug 2020)**
 - Led initiatives to improve the research environment and address research scholars' issues on campus.
 - Developed and managed the RSF IIT Bombay website to enhance communication and resource sharing.
 - Organized multiple research events and workshops as a core council member, fostering an innovative and collaborative community.
- **System Administrator, Hostel 12 – IIT Bombay (Apr 2019 – Mar 2020)**
 - Maintained and upgraded hostel IT infrastructure; resolved network and computer issues for 500+ residents.
 - Coordinated technical events and workshops in the hostel, improving digital connectivity and technical awareness among students.

- **Web Secretary, Dept. of Energy Science & Engineering, IIT Bombay** (Apr 2018 – Mar 2019)
 - Oversaw content management and regular updates of the departmental website.
 - Implemented improvements to site structure and design, ensuring a user-friendly interface and up-to-date information for students and faculty.
- **Web & Media Manager, Energy Day (Institute Event) – IIT Bombay** (Apr 2018)
 - Developed the “Energy Day 2018” event website and managed social media outreach to promote sustainable energy awareness.
 - Coordinated with student volunteers and vendors to ensure successful event publicity and on-site execution.
- **Event Manager, Int’l Conference on Advances in Energy Research (ICAER 2019) – IIT Bombay** (Dec 2019)
 - Designed conference promotional materials and managed venue setup/logistics for this international event.
 - Liaised with organizing committees and vendors to meet technical requirements and ensure smooth operations during the conference.
- **Event Coordinator, Int’l Conference on Advances in Energy Research (ICAER 2017) – IIT Bombay** (Dec 2017)
 - Assembled and proofread conference abstract booklets; assisted with venue management and participant coordination.
- **Event Coordinator, Abhyuday Social Festival – IIT Bombay** (Feb 2016)
 - Promoted event activities in local schools to boost community participation.
 - Assisted in event logistics and volunteer management for this campus social initiative.

Publications and Presentations

Journal Articles (Under Review / Preparation):

- Ali, A., Ramadesigan, V., & Monder, D. S. (2025). Modelling and simulation based impact analysis of electrode parameters on the performance of vanadium redox flow batteries. *Journal of Power Sources*, 653, 237659. <https://doi.org/10.1016/j.jpowsour.2025.237659>
- Abdul Ali & V. Ramadesigan (2025). “Singular perturbation-based initialization and one-step solution approach for index-n differential-algebraic equation systems.” *Mathematics and Computers in Simulation* (under review).
- Abdul Ali & V. Ramadesigan (2025). *Impact of electrode structure on ion crossover and vanadium redox flow battery performance*. (Manuscript in preparation)

Conference Presentations:

- Ali, A.; Ramadesigan, V.; Saha, P. (2022). “Impact of Electrode Parameters on the Crossover of Vanadium Ions Across the Membrane in a VRFB System.” 73rd Annual Meeting of the International Society of Electrochemistry (ISE 2022 – Virtual).
- Ali, A.; Ramadesigan, V. (2023). “A Comparative Study of Electrode Parameters of Vanadium Redox Flow Batteries for Improved Design and Performance.” *Electrochemical Society Meeting Abstracts*, MA2023-02(59), Abstract #2882 (ECS Meeting 2023). (Presented at the 244th Electrochemical Society Meeting, Gothenburg, Sweden)

- Saha, P.; Ali, A.; Ramadesigan, V. (2023). “Estimation of State-of-Charge and Energy Efficiency of Supercapacitors Using a 1-D Electrochemical Model.” *Electrochemical Society Meeting Abstracts*, MA2023-02(1), Abstract #4 (ECS Meeting 2023). (Presented at the 244th Electrochemical Society Meeting, Gothenburg, Sweden)

Entrepreneurial Experience

- **Founder, eRentals HND Pvt Ltd, Mumbai, India** (Operational since January 2023): Founded and scaled a rental services startup, achieving revenue exceeding ₹2 Crore.
- **Startup and Innovation Training, Indian Institute of Technology Bombay, Mumbai, India** (2017 – 2018): Completed entrepreneurship courses and engaged in innovation-driven projects, including Proof of Concept Development and Smart Campus initiatives